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QSS

QUARTERLY EXAMINATION - 2025

12 - Std MATHEMATICS

Time : 3.00 hrs. Marks : 90

PART – A

I CHOOSE THE CORRET ANSWER :- 20 X 1 = 20

1) If $P=\begin{bmatrix}1 & x & 0\\ 1 & 3 & 0\\ 2 & 4 & -2\end{bmatrix}$ is the adjoint of 3×3 matrix A and $|A|=4$, then x is

(a) 15 (b) 12 (c) 14 (d) 11

2) If $A=\begin{bmatrix}\frac{3}{5} & \frac{4}{5}\\ x & \frac{3}{5}\end{bmatrix}$ and $A^T=A^{-1}$, then the value of x is

(a) $\frac{-4}{5}$ (b) $\frac{-3}{5}$ (c) $\frac{3}{5}$ (d) $\frac{4}{5}$

3) If $\left|z-\frac{3}{z}\right|=2$ then the least value of $|z|$ is

(a) 1 (b) 2 (c) 3 (d) 5

4) If $\frac{z-1}{z+1}$ is purely imaginary, then $|z|$ is

(a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 3

5) If α and β are the roots of $x^2+x+1=0$, then $\alpha^{2020}+\beta^{2020}$ is

(a) -2 (b) -1 (c) 1 (d) 2

6) A zero of x^3+64 is

(a) 0 (b) 4 (c) $4i$ (d) -4

7) The number of positive roots of the polynomial $\sum_{r=0}^{n} n_{c_r}(-1)^r x^r$ is

(a) 0 (b) n (c) $< n$ (d) r

8) $\sin^{-1}\frac{3}{5}-\cos^{-1}\frac{12}{13}+\sec^{-1}\frac{5}{3}-\operatorname{cosec}^{-1}\frac{13}{12}$ is equal to

(a) 2π (b) π (c) 0 (d) $\tan^{-1}\frac{12}{65}$

9) If $\cot^{-1}x=\frac{2\pi}{5}$ for some $x\in R$, the value of $\tan^{-1}x$ is

(a) $-\frac{\pi}{10}$ (b) $\frac{\pi}{5}$ (c) $\frac{\pi}{10}$ (d) $-\frac{\pi}{5}$

10) The equation $\tan^{-1}x-\cot^{-1}x=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ has

(a) no solution (b) unique solution (c) two solutions (d) infinite number of solutions

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27. Find the equation of the ellipse with foci $(0, \pm 4)$ and end points of major axis are $(0, \pm 5)$.

28. Find the distance between the parallel planes $x + 2y - 2z + 1 = 0$ and $2x + 4y - 4z + 5 = 0$.

29. The volume of the parallelopiped whose coterminus edges are $7\hat{i} + \lambda\hat{j} - 3\hat{k}, \hat{i} + 2\hat{j} - \hat{k}, -3\hat{i} + 7\hat{j} + 5\hat{k}$ is 90 cubic units. Find the value of λ.

30. Find the equation of a point which moves so that the sum of its distance from (-4,0) and (4,0) is 10.

PART – C

III ANSWER ANY SEVEN QUESTIONS (Q.NO : 40 IS COMPULSORY) :- 7 X 3 = 21

31. Find the rank of the matrix $\begin{bmatrix} 2 & -2 & 4 & 3 \\ -3 & 4 & -2 & -1 \\ 6 & 2 & -1 & 7 \end{bmatrix}$ by reducing it to an Echelon form.

32. Solve the following system of linear equations by matrix inversion method:
$2x - y = 8, \quad 3x + 2y = -2$

33. Show that $\left(\frac{19+9i}{5-3i}\right)^{15} - \left(\frac{8+i}{1+2i}\right)^{15}$ is purely imaginary.

34. Solve the equation $9x^3 - 36x^2 + 44x - 16 = 0$ if the roots form an arithmetic progression.

35. Find the domain of (i) $f(x) = \sin^{-1}\left(\frac{|x|-2}{3}\right) + \cos^{-1}\left(\frac{1-|x|}{4}\right)$.

36. Find the value of $\sin^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \cot^{-1}(2)$

37. If the normal at the point 't_1' on the parabola $y^2 = 4ax$ meets the parabola again at the point 't_2' then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

38. If the equation $3x^2 + (3-p)xy + qy^2 - 2px = 8pq$ represents a circle, find p and q. Also determine the centre and radius of the circle.

39. Find the point of intersection of the line $x - 1 = \frac{y}{2} = z + 1$ with the plane $2x - y + 2z = 2$. Also, find the angle between the line and the plane.

40. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = 0$ and the angle between $\vec{a}$ and $\vec{c}$ and is $\frac{\pi}{3}$. Prove that $\vec{b} = \pm\frac{2}{\sqrt{3}}(\vec{a} \times \vec{b})$.

PART – D

IV ANSWER ALL THE QUESTIONS :- 7 X 5 = 35

41 a) Solve the following system of linear equations by Cramer's rule:
$\frac{3}{x} - \frac{4}{y} - \frac{2}{z} - 1 = 0, \quad \frac{1}{x} + \frac{2}{y} + \frac{1}{z} - 2 = 0, \quad \frac{2}{x} - \frac{5}{y} - \frac{4}{z} + 1 = 0$ **(OR)**

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11) The centre of the circle inscribed in a square formed by the lines $x^2 - 8x - 12 = 0$ and $y^2 - 14y + 45 = 0$ is
(a) (4,7) (b) (7,4) (c) (9,4) (d) (4,9)

12) If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is
(a) 3 (b) -1 (c) 1 (d) 9

13) Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
(a) $2ab$ (b) ab (c) $\sqrt{ab}$ (d) $\frac{a}{b}$

14) If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}$ is perpendicular to $\vec{b}$, and is parallel to $\vec{c}$ then $\vec{a} \times (\vec{b} \times \vec{c})$ is equal to
(a) $\vec{a}$ (b) $\vec{b}$ (c) $\vec{c}$ (d) $\vec{0}$

15) The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

16) If the planes $\vec{r} \cdot (2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ are parallel, then the value of λ and μ are
(a) $\frac{1}{2}, -2$ (b) $-\frac{1}{2}, 2$ (c) $-\frac{1}{2}, -2$ (d) $\frac{1}{2}, 2$

17) If $|A| = 2$ and A is 2 X 2 matrix, then the value of $|\text{adj}\{\text{adj}(\text{adj } A^2)\}|$ is
(a) 4 (b) 16 (c) 64 (d) 128

18) If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$ then xy + yz + zx is
(a) 1 (b) 0 (c) -3 (d) 3

19) If the line $2x + 3y = 12$ touches the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 2$ at the point is
(a) (3,2) (b) (2,3) (c) $(-3,-2)$ (d) (3,4)

20) The image of the point (3,-1,11) in the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ is
(a) (1,11,3) (b) (11,11,13) (c) (3,1,11) (d) (1,3,11)

PART – B

II ANSWER ANY SEVEN QUESTIONS (Q.NO : 30 IS COMPULSORY) :- 7 X 2 =14

21. Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.

22. Find the square root of $6 - 8i$.

23. Find the value of $\sum_{k=1}^{8} \left(\cos\frac{2k\pi}{9} + i\sin\frac{2k\pi}{9} \right)$.

24. Find a polynomial equation of minimum degree with rational coefficients, having $2 - \sqrt{3}$ as a root.

25. Find the principal value of $\sin^{-1}\left(\sin\left(\frac{5\pi}{4} \right) \right)$.

26. Prove that $\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}$ for $|x| < 1$.

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b) If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}, \vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}, \vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$, verify that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a}.\vec{c})\vec{b} - (\vec{b}.\vec{c})\vec{a}$.

42. a) If $z = x + iy$ is a complex number such that $Im\left(\frac{2z+1}{iz+1}\right) = 0$, show that $2x^2 + 2y^2 + x - 2y = 0$. **(OR)**

b. If the straight lines $\frac{x-1}{2} = \frac{y+1}{\lambda} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{\lambda}$ are coplanar, find λ and Cartesian equation of the plane containing these two lines.

43. a. Find the number of solutions of the equation $\tan^{-1}(x-1) + \tan^{-1}x + \tan^{-1}(x+1) = \tan^{-1}3x$ **(OR)**

b. Investigate the values of λ and μ the system of linear equations $2x + 3y + 5z = 9$, $7x + 3y - 5z = 8$, $2x + 3y + \lambda z = \mu$, have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

44. a. Find the equation of the circle passing through the points $(1,1), (2,-1)$, and $(3,2)$.

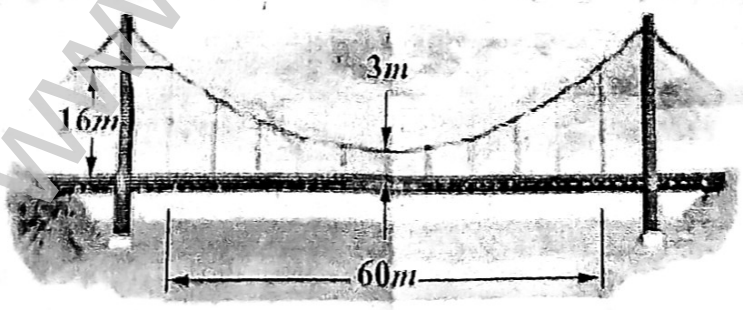
(OR) b. Solve the following equation: $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$.

45. a. Identify the type of conic and find centre, foci, vertices, and directrices of the conic $9x^2 - y^2 - 36x - 6y + 18 = 0$. **(OR)**

b. Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point $(2,3,6)$ and parallel to the straight lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-3}{1}$ and $\frac{x+3}{2} = \frac{y-3}{-5} = \frac{z+1}{-3}$

46. a. Find the value of $\cos\left(\sin^{-1}\left(\frac{4}{5}\right) - \tan^{-1}\left(\frac{3}{4}\right)\right)$. **(OR)**

b. Parabolic cable of 60m portion of the roadbed of a suspension bridge are positioned as shown below. Vertical cables are to be spaced every 6m along this portion of the roadbed. Calculate the lengths of first two of these vertical cables from the vertex.



47. a. Find all the values of $\left(\sqrt{3} - i\right)^{\frac{2}{3}}$. **(OR)**

b. Solve : $\sin^{-1}\left(\frac{x}{5}\right) + cosec^{-1}\left(\frac{5}{4}\right) - \tan^{-1}\left(\frac{2}{5}\right) = \cot^{-1}\left(\frac{2}{5}\right)$.

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